CPE462 - VHDL: Simulation and Synthesis - Fall'11

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Homework Assignment #18

Due Date: Monday, December 12 2011

Question 1 - Write a PROCEDURE that receives eight signed values and returns their average, the largest value, and the lowest value. Call the return values ave, max, and min. Place your procedure in a package. Then write an application with a call to it in order to test its functionality.

Question 2 - Overloaded "not" Operator: The NOT operator allows the inversion of binary values. For example, if x="1000" is a STD_LOGIC_VECTOR value, then NOT x could be used, producing "0111". However, if x had been declared as an INTEGER, such operation would not be allowed. Write a "not" function capable of inverting integers.

Question 3 - Write a function capable of logically shifting a STD_LOGIC_ VECTOR signal to the left by a specified amount. Two arguments must be passed to the function: the value to be shifted (STD_LOGIC_VECTOR), plus a NATURAL value specifying the amount of shift. Place your function in a package. Then write an application with a call to your function in order to test it.